

HAT-P-32b

R-0153

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_231111 · created 2026-07-16T14:25:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460259.8687 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1487 +/- 0.0049
Transit depth (Rp/Rs)^2	2.21 +/- 0.15 [%]
Orbital Inclination (inc)	86.22 +/- 1.34
Airmass coefficient 1 (a1)	1.2338 +/- 0.0072
Airmass coefficient 2 (a2)	-0.1371 +/- 0.0029
Scatter in the residuals of the lightcurve fit is	0.57 %
Best Comparison Star	None
Optimal Aperture	4.51
Optimal Annulus	12.09
Transit Duration (day)	0.1254 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1487 ± 0.0049 vs 0.1489 (deviation 0.04σ)
Duration fit vs published	0.1254 d vs 0.1304 d (deviation 1.35σ)
Mid-time O–C	-0.6 min
Depth SNR	29.5
Residual scatter	0.57 %

Light curve

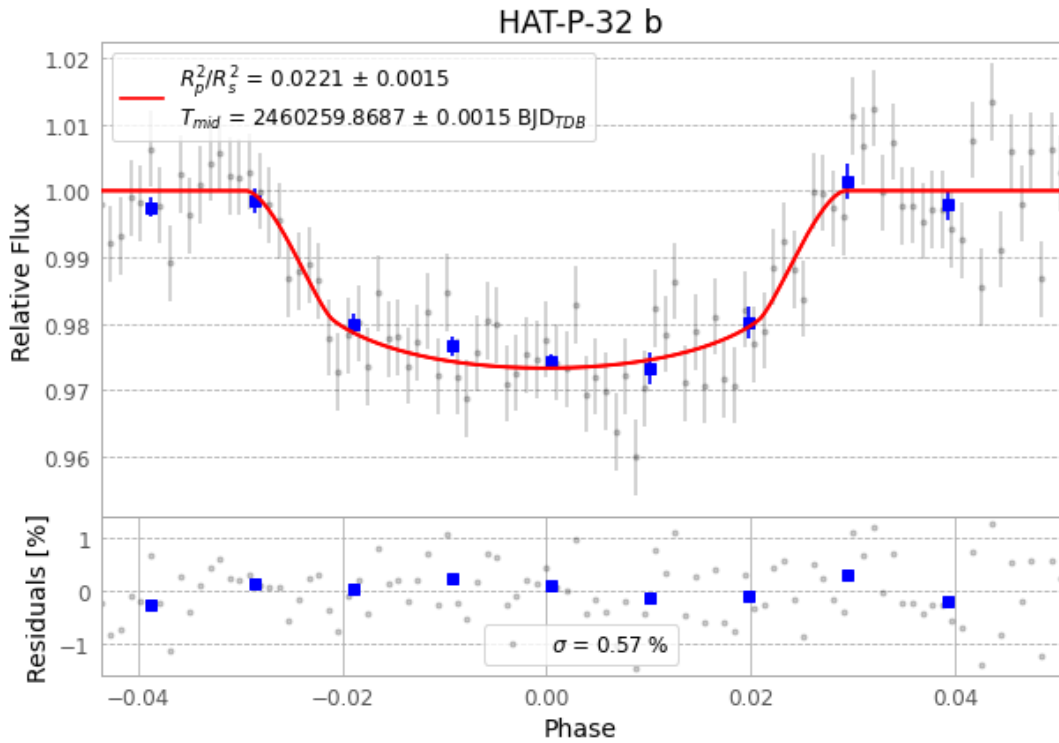


Figure 1 — FinalLightCurve_HAT-P-32 b_2023-11-10.png (SHA-256 4fde0445805c01d2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3e842c2f5c00a2eb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 6eeacf9

Data provenance

98 FITS frames (64 MB), HATP-32231111062716.FITS → HATP-3223111111816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-231111.log (41331fbc0135...), FinalLightCurve_HAT-P-32 b_2023-11-10.png (4fde0445805c...), FinalLightCurve_HAT-P-32 b_2023-11-10.csv (91c8e53451e4...)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 14:25Z.